

AN ASPIRIN A DAY TO PREVENT PREMATURITY

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Preterm deliveries are the leading cause of perinatal mortality and morbidity in the United States.¹ Most of the perinatal mortality and morbidity are associated with deliveries occurring prior to 30 weeks' gestation.² Infants delivered prior to 30 weeks' gestation are at increased risk for severe respiratory distress syndrome, intraventricular hemorrhage, necrotizing enterocolitis, sepsis, and growth retardation.⁹ In addition, these infants are at increased risk for long-term mortality and morbidity (neurologic deficits, chronic lung disease, and blindness).⁹ The majority of preterm deliveries are usually due to spontaneous preterm labor and/or preterm premature rupture of the membranes; however, in about one fourth of the cases, preterm delivery is usually performed for maternal or fetal reasons, particularly in pregnancies complicated by uteroplacental vascular insufficiency or umbilical placental vascular insufficiency.¹⁰ Recently, the use of low-dose aspirin (60 to 150 mg/d) has been shown to improve pregnancy outcome in women considered at risk for uteroplacental or umbilical placental vascular insufficiency.²⁶ This article reviews the safety and clinical efficacy of low-dose aspirin in preventing preterm birth, fetal growth retardation, and poor perinatal outcome.

ACTIONS OF REGULAR AND LOW-DOSE ASPIRIN

The major effects of aspirin are related to inhibition of prostaglandins synthesis in various body tissues. Prostaglandins are the biologically active metabolites of arachidonic acid. Arachidonic acid is liberated from mem-

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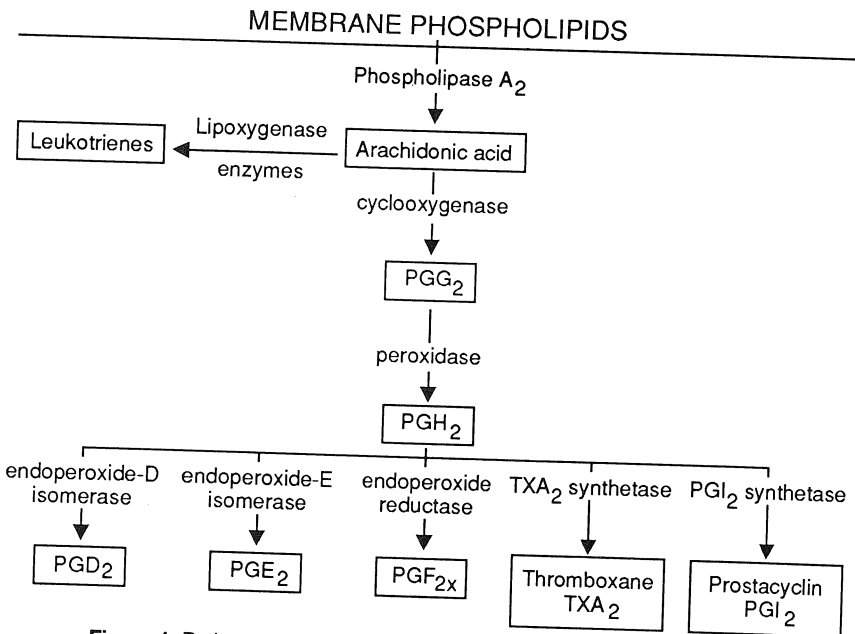


Figure 1. Pathways leading to the production of various prostaglandins.

brane phospholipids by way of the action of phospholipase A₂. It is then converted by the cyclo-oxygenase pathway into prostaglandin G₂ (PGG₂), which serves as the precursor to the various biologically active prostaglandins (Fig. 1).

Aspirin inhibits the synthesis of the cyclic endoperoxide PGG₂ by irreversibly acetylating and inactivating cyclo-oxygenase. These effects in turn will result in simultaneous inhibition of the synthesis of four active substances (PGE₂, PGF_{2α}, prostacyclin, and thromboxane) that have major biologic actions during pregnancy (Table 1). It is important to note that this

Table 1. BIOLOGIC ACTIONS OF PROSTAGLANDINS AND THROMBOXANE IN PREGNANCY

	Thromboxane TXA ₂	Prostacyclin PGI ₂	PGE ₂	PGF _{2α}
Vasoconstriction	↑↑	↓	↓	↑↑
Uterine activity	↑↑	↓	↓	↑↑
Bronchial tone	↑↑	↓	↓	↑↑
Platelet aggregation	↑	↓	↓	↑
Ductus arteriosus patency	—	↑	—	—
Renal blood flow*	?	↑	↑	—

* Maternal, fetal, and neonatal.

nonselective inhibition of the various prostaglandins and thromboxane is usually seen following chronic use of regular doses of aspirin (> 325 mg/d).

Acetylsalicylic acid (ASA) is the chemical ingredient of aspirin. It is a weak acid and therefore exists primarily in the ionized form in blood or body tissues. Following oral ingestion, aspirin is rapidly absorbed from the stomach and upper intestine.⁴ It is then hydrolyzed to salicylic acid by intestinal, hepatic, and plasma esterases. The acetyl moiety in ASA is the active component that inhibits the cyclo-oxygenase enzymes in various cells. An adult tablet of aspirin contains ≥ 325 mg ASA, whereas a baby aspirin contains 81 mg ASA; however, the systemic bioavailability of aspirin is constant after single oral doses ranging from 20 to 1300 mg. In addition, systemic bioavailability is similar after a single dose and after chronic oral administration of 325 mg of aspirin.¹⁵

Aspirin inactivates the cyclo-oxygenase enzymes in both the platelets and vascular endothelium. In vitro studies suggest that platelet cyclo-oxygenase is more sensitive to inhibition by very low doses of aspirin (< 80 mg) than vascular endothelial cyclo-oxygenase.¹⁸ Pedersen and Fitzgerald¹⁵ investigated the dose-related kinetics of low-dose aspirin in non-pregnant subjects with the use of deuterated aspirin analog. They found that platelets passing through the gut capillaries were exposed to significantly higher concentrations of aspirin than the platelets in the peripheral circulation. They concluded that low-dose aspirin causes relatively high concentrations of the drug in the portal circulation leading to total inhibition of the cyclo-oxygenase enzymes in the platelets passing through the gut capillaries, whereas the concentration of aspirin in the peripheral circulation remains too low to affect the enzyme in the vascular endothelium. This biochemical selectivity of platelet thromboxane production by low-dose aspirin has also been reported during pregnancy.¹⁸ Indeed, several recent studies describing the use of low-dose aspirin during pregnancy found that aspirin alters the balance between prostacyclin and thromboxane in favor of prostacyclin.^{6,7,18,25}

The prostacyclin/thromboxane balance may be important in regulating the uteroplacental and umbilical placental circulations, with an imbalance in this ratio leading to uteroplacental and/or umbilical placental vascular insufficiency and even pathologic vascular occlusive disease.²⁶ In pregnancies complicated by preeclampsia, fetal growth retardation, vascular disease, anticardiolipin antibodies, lupus anticoagulant antibodies, idiopathic thrombocytopenia, and thrombotic thrombocytopenic purpura, the predominant prostanoid produced may be thromboxane. Therefore, treatment with low-dose aspirin has been used to improve pregnancy outcome in women with or at risk for the above complications.

PREVENTION OF PRETERM BIRTH

There are two studies describing the use of pharmacologic doses of aspirin (2 to 3.6 g/d) to arrest spontaneous preterm labor.^{12,28} Both studies found a beneficial effect; however, the number of patients studied in these

reports were too small to draw any conclusions. In addition, neither of these studies included a control group. Moreover, there is no evidence from the published randomized trials that low-dose aspirin reduces the incidence of spontaneous preterm birth⁴; however, there is definite evidence from the randomized trials that low-dose aspirin reduces the incidence of preterm deliveries due to preeclampsia or fetal growth retardation.

PREVENTION OF PREECLAMPSIA

In pregnancies complicated by preeclampsia, there is an increased risk of perinatal mortality and morbidity, mainly related to preterm delivery, fetal growth retardation, uteroplacental insufficiency, and abruptio placentae.^{14,17} These risks are particularly increased in women developing severe preeclampsia prior to 34 weeks' gestation and more so in women developing severe preeclampsia in the second trimester.^{14,17} Therefore, prevention of early onset preeclampsia (<37 weeks' gestation) results in significant reduction in preterm delivery as a consequence of this complication.

There are several randomized-controlled studies describing the use of low-dose aspirin (60 to 150 mg/d) to prevent preeclampsia.^{3,5,6,13,16,22,24} The women enrolled in these studies were selected based on the presence of one or more of the risk factors listed in Table 2. A total of 285 women received either a placebo or no treatment, and 370 women received aspirin with or without dipyridamole. Pregnancy outcome in the studied women is depicted in Figure 2. The treated women had significantly lower incidences of both preeclampsia ($P < 0.00$) and fetal growth retardation ($P < 0.00$).

Beaufils et al⁵ studied 93 high-risk women: 48 were treated with aspirin 150 mg/d plus dipyridamole 300 mg/d from 12 weeks' gestation until delivery, and 45 received no treatment. They reported six cases (13%) of preeclampsia, five fetal deaths (11%), and three cases of abruptio placentae (11%) in the control group, but no such complication in the treated group. In addition, treated women had significantly higher mean gestational age at time of delivery (38 versus 36 weeks, $P < 0.001$), larger neonatal birth

Table 2. FACTORS USED TO IDENTIFY WOMEN AT RISK FOR PREECLAMPSIA

Preeclampsia in previous pregnancies
Chronic hypertension/renal disease
Poor outcome in previous pregnancies
Repetitive pregnancy loss
Fetal demise
Fetal growth retardation
Abruptio placentae
Angiotensin-II sensitivity at 28 weeks
Positive roll-over test at 28 weeks
Abnormal uterine Doppler studies at 18 and 24 weeks

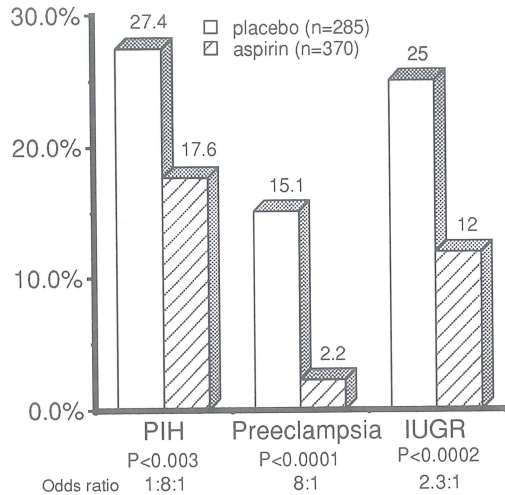


Figure 2. Randomized trials of low-dose aspirin in pregnancies at risk for preeclampsia.

weights (3172 versus 2625 g, $P < 0.001$), and significantly higher mean placental weights (599 versus 509 g, $P < 0.01$). The incidence of fetal growth retardation was significantly lower in the treated group (8.3% versus 28.8%, $P < 0.005$), and all seven cases of severe fetal growth retardation occurred in the control group.

Wallenburg et al²⁴ studied 44 primigravidae who had increased sensitivity to angiotensin-II infusions at 28 weeks' gestation. Twenty-one women received aspirin 60 mg/day, and the other 23 women received a matched placebo. There were seven cases of preeclampsia, one case of eclampsia, and three deliveries prior to 37 weeks in the placebo-treated group, but none in the aspirin-treated group. In addition, the incidence of fetal growth retardation was significantly lower in the aspirin-treated group (19% versus 39%).

Schiff and associates¹⁶ studied pregnancy outcome in 65 women who had a positive roll-over test at 28 to 29 weeks' gestation. Thirty-four women received aspirin 100 mg/d, and the other 31 received a matched placebo. The incidence of preeclampsia was significantly lower in the aspirin-treated group (2.9% versus 22.6%, $P < 0.02$). There were no perinatal deaths in either group; however, the aspirin-treated group had lower incidence of preterm deliveries and lower incidences of neonatal complications (Table 3).

Benigni and associates⁶ studied pregnancy outcome in 33 high-risk women who were randomly assigned to receive 60 mg of aspirin ($n = 17$) or placebo ($n = 16$) daily from the 12th week of gestation until delivery. There were no cases of preeclampsia or fetal death in either group; however the aspirin-treated group had lower incidence of fetal growth retardation, longer gestational ages, and larger neonatal birth weights (Table 4).

McParland et al¹³ studied pregnancy outcome in 100 nulliparous women who were identified to have persistently abnormal Doppler utero-

Table 3. PERINATAL OUTCOME IN WOMEN RECEIVING ASPIRIN OR A PLACEBO TO PREVENT PREECLAMPSIA

	Placebo n = 31*	Aspirin n = 34
GA at delivery (wk)†	37.3 ± 0.9	38.9 ± 1.1
< 37 weeks # (%)	6 (19.4)	2 (5.9)
Birth weight (g)	2706	3037
< 10th percentile # (%)	6 (19.4)	2 (5.9)
Adjusted birth weight percentile‡	38.6 ± 12.2	43.9 ± 13.7
Admitted to NICU	7 (21.9)	2 (5.9)
Requiring ventilatory support	3 (9.4)	1 (3.0)

* 32 infants (one set of twins)

† P < 0.001

‡ P < 0.01

Adapted from Schiff E, Peleg E, Goldenberg M, et al: The use of aspirin to prevent pregnancy-induced hypertension and lower the ratio of thromboxane A₂ to prostacyclin in relatively high risk pregnancies. N Engl J Med 321:351–356, 1989; with permission.

placental resistant indices at 18 and 24 weeks' gestation. These women were randomized to receive either 75 mg of aspirin daily (n = 48) or a placebo (n = 52). Hypertension arising before 37 weeks' gestation developed in 17% of the placebo group but in none of the aspirin group, and the incidence of preeclampsia was significantly higher in the placebo group. There were no differences between the two groups regarding gestational age at delivery, neonatal birth weights, or in the incidence of fetal growth retardation. There were, however, higher incidences of birth weights below 2500 g and below 1500 g in the placebo group (Table 5).

Azar and Turpin³ studied pregnancy outcome in 91 high-risk women who were randomly assigned to receive 100 mg of aspirin plus 300 mg dipyridamole (n = 46) or no treatment (n = 45) daily from the 16th week of gestation until delivery. The authors found that treated women had signifi-

Table 4. PREGNANCY OUTCOME WITH ASPIRIN TO PREVENT PREECLAMPSIA

	Placebo n = 16	Aspirin n = 17
GA at delivery (wk)	35 ± 4	39 ± 2
Birth weight (g)	2264 ± 1072	2922 ± 599
< 10th percentile # (%)	6 (37.5)	2 (11.8)
PIH # (%)	3 (17.8)	0

PIH = pregnancy-induced hypertension.

Adapted from Benigni A, Gregorini G, Frusca T, et al: Effect of low dose aspirin on fetal and maternal generation of thromboxane by platelets in women at risk for pregnancy-induced hypertension. N Engl J Med 321:357–362, 1989; with permission.

Table 5. DOPPLER ULTRASOUND AND ASPIRIN IN RECOGNITION AND PREVENTION OF PREGNANCY-INDUCED HYPERTENSION

	Placebo n = 52	Aspirin n = 48
GA at delivery (wk)	38.7 ± 3.9	39.5 ± 2.1
Birth weight (g)	2954 ± 852	3068 ± 555
<2500 g	13 (25)	7 (15)
<1500 g	4 (8)	0
<5th percentile	7 (14)	7 (14)
perinatal deaths	3 (5.8)	1 (2.1)
preeclampsia†	10 (19)	1 (2.1)

** All three related to severe preeclampsia in second trimester.

† $P < 0.02$

Results expressed as either mean + SD or numbers and percent.

Adapted from McParland P, Pearce JM, Chamberlain GVP: Doppler ultrasound and aspirin in recognition and prevention of pregnancy-induced hypertension. *Lancet* 335:1552–1555, 1990; with permission.

cantly longer duration of pregnancy (37.9 ± 1.2 versus 35.4 ± 1.1 weeks, $P < 0.01$), and lower incidence of preeclampsia (2.2% versus 8.9%, $P < 0.05$). In addition, there were two fetal deaths in each group.

Recently, Uzan et al²² reported the results of a multicenter trial in 229 high-risk women who were treated with either a placebo ($n = 73$), aspirin 150 mg/d ($n = 81$), or aspirin 150 mg/d plus dipyridamole 225 mg/d ($n = 75$). The authors reported a lower incidence of preeclampsia (3% versus 11%, $P < 0.05$) and a better perinatal outcome in the combined treatment groups. Further details of this trial will be described below under prevention of fetal growth retardation.

PREVENTION OF FETAL GROWTH RETARDATION

Intrauterine fetal growth retardation is an important cause of preterm birth, and it is a major cause of neonatal morbidity. Several recent studies suggest that low-dose aspirin prevents fetal growth retardation and thus preterm delivery in women at-risk for this complication. Wallenburg and Rotmans²⁶ studied pregnancy outcome in 24 multigravid women with a history of at least two previous pregnancies complicated by fetal growth retardation and placental infarction. The women were treated with 1 to 1.6 mg/kg of aspirin plus 225 mg dipyridamole daily from 16 to 34 weeks' gestation in a total of 30 pregnancies. Results obtained in this group were compared with those in 24 multigravid women with similar obstetric histories who were followed through 27 subsequent pregnancies. These latter women received comparable antenatal care, but they did not receive antiplatelet therapy. The authors reported lower incidence of fetal growth retardation and absent severe fetal growth retardation in the treated women. This study was subsequently expanded to include a total of 44 treated pregnancies in 37 women.²⁷ Results of both studies are summarized in

Table 6. PREVENTION OF RECURRENT FETAL GROWTH RETARDATION BY LOW-DOSE ASPIRIN AND DIPYRIDAMOLE*

	Control n = 24	Aspirin and Dipyridamole n = 37
No. of pregnancies	27	44
spontaneous abortion	1	2
fetal death	0	3
GA at delivery (wk)	38	37
< 5th percentile*	12 (46.2)	3 (7.1)
< 10th percentile†	16 (61.5)	5 (11.9)
Respiratory distress syndrome	3	5

* $P < 0.0003$ † $P < 0.0001$

Adapted from Wallenburg HCS, Rotmans N: Effect of low-dose aspirin on vascular refractoriness in angiotensin-sensitive primigravid women. *Am J Obstet Gynecol* 157:1230-1235, 1987; and Wallenburg HCS, Rotmans N: Prevention of recurrent fetal growth retardation. *Lancet* 1:939, 1988; with permission.

Table 6. The authors attributed the improved perinatal outcome in treated women to low-dose aspirin; the effects were thought to be related to correction of the prostacyclin-thromboxane imbalance in the uteroplacental vasculature.

Trudinger et al²⁰ studied 46 women who had abnormal umbilical artery flow velocity systolic/diastolic ratios at 28 to 36 weeks' gestation. These women were then randomized to receive either aspirin 150 mg/d (n = 22) or a placebo (n = 24). At time of randomization, 34 women had a high systolic/diastolic ratio (>95th but <99.5th percentile), and 12 women had an extremely high systolic/diastolic ratio (>99.5th percentile). Aspirin therapy was associated with an improved neonatal outcome in the women who had a high initial umbilical artery systolic/diastolic ratio (Table 7). The absence of benefit from aspirin in women with extreme ratio was considered to indicate that irreversible vascular sclerosis had already developed in the fetal umbilical placental circulation of these women. The authors suggested that the beneficial effects of aspirin in pregnancies complicated by umbilical placental insufficiency are related to inhibition of thrombosis in the fetal umbilical placental circulation. In a subsequent report,²¹ the same authors conducted a similar trial in 27 women with twin pregnancies in which there was no evidence of placental insufficiency at 28 to 30 weeks' gestation. These women were randomized to either aspirin 100 mg/day or placebo (Table 8). There were no differences between the two groups in either placental weight or neonatal outcome; however, the results of this study are not conclusive considering the limited sample size.

Uzan et al²² reported a multicenter, placebo-controlled, double-blind trial in 323 women at 15 to 18 weeks' gestation who were selected on the basis of fetal growth retardation and/or fetal death or abruptio placentae in at least one previous pregnancy. The trial was conducted in two phases; in the first phase, there were three groups: placebo (n = 74), aspirin 150 mg/d

Table 7. LOW-DOSE ASPIRIN THERAPY IMPROVES FETAL WEIGHT IN UMBILICAL PLACENTAL INSUFFICIENCY

	High Umbilical S/D Ratio*		Extreme Umbilical S/D Ratio†	
	Placebo n = 20	Aspirin n = 14	Placebo n = 4	Aspirin n = 8
GA at randomization (wk)	31.6	31.9	31.9	31.3
GA at delivery (wk)	35.9	37.0	33.0	33.3
Birth weight (g)‡	2035	2551	1558	1393
Percentile birth weight (%)‡	21.5	43.1	18.8	17.5
Neonatal head circumference (%)‡	31.2	32.9	29.3	28.5
Placental weight (g)‡	399.4	535.4	335.0	294.6

* Umbilical systolic/diastolic ratio > 95th but < 99.95th percentile

† Umbilical systolic/diastolic ratio > 99.95th percentile

‡ $P < 0.025$ for the high umbilical S/D ratio group only

Adapted from Trudinger BJ, Cook CM, Thompson R, et al: Low-dose aspirin therapy improves fetal weight in umbilical placental insufficiency. *Am J Obstet Gynecol* 159:681–685, 1988; with permission.

($n = 81$), and aspirin 150 mg/d plus dipyridamole 225 mg/d ($n = 75$). The second phase had two groups: aspirin ($n = 47$) and dipyridamole ($n = 46$). The results of the study were divided into two double-blind trials: trial A (treatment versus placebo) and trial B (aspirin versus aspirin plus dipyridamole). Table 9 compares outcome of pregnancies in trial A. Treated women had significantly higher mean birth weight and lower neonatal complications than the placebo-treated women. In addition, in women with two or more poor outcomes in previous pregnancies, the treated group had significantly lower incidence of fetal growth retardation (13% versus 42%, $p < 0.01$), and a borderline significant difference for fetal death and abruptio placentae (2% versus 12%, $P = 0.054$). On the other hand, in trial B, there were no differences between women receiving aspirin ($n = 127$) and those receiving aspirin plus dipyridamole ($n = 119$) regarding either birth weight or neonatal outcome.

Elder and associates¹¹ studied pregnancy outcome in 38 multigravid

Table 8. LOW-DOSE ASPIRIN AND TWIN PREGNANCY*

	Placebo n = 12	Aspirin (100 mg/d) n = 15
GA at randomization (wk)	28.9	28.5
GA at delivery (wk)	36.2	36.2
Birth weight (g)	2466	2524
Placental weight (g)	1053	1044

* Included twin pregnancies with absent evidence of placental insufficiency.

Adapted from Trudinger BJ, Cook CM, Giles WB, et al: Low-dose and twin pregnancy. *Lancet* 2:1214, 1989; with permission.

Table 9. PREVENTION OF FETAL GROWTH RETARDATION WITH LOW-DOSE ASPIRIN: FINDINGS OF THE EPREDA TRIAL

	Placebo n = 73 (%)	Aspirin n = 156 (%)
GA at randomization (wk)	16.3 + 2	16.3 + 1.4
GA at delivery (wk)	36.9 + 3.9	37.7 + 2.7
Birth weight (g)**	2526 + 848	2751 + 670
< 10th percentile	19 (26)	20 (13)
Perinatal deaths	6 (8.2)	7 (4.5)
Abruptio placentae	6 (8.2)	7 (4.5)
Admission to NICU	24 (34.8)	41 (26.6)
Requiring artificial ventilation	10 (14.5)	13 (8.4)

* $P = 0.05$
† $P = 0.03$
GA = gestational age; NICU = neonatal intensive care unit
Adapted from Uzan S, Beaufils M, Breart G, et al: Prevention of fetal growth retardation with low-dose aspirin: Findings of the EPREDA trial. Lancet 337:1427-1431, 1991; with permission.

women with very bad obstetric histories, largely due to severe fetal growth retardation. Sixteen women had systemic lupus erythematosus, and the other women had previous histories of severe pregnancy-induced hypertension or severe fetal growth retardation. These women were treated with 75 mg of aspirin daily from the first or second trimester until delivery. Table 10 summarizes pregnancy outcome in these women before and after using low-dose aspirin. The data demonstrate a significant improvement in perinatal outcome following the use of low-dose aspirin.

Table 10. LOW-DOSE ASPIRIN IN WOMEN WITH VERY BAD OBSTETRIC HISTORIES*

	Previous Pregnancies n = 62 (%)	Subsequent Pregnancies with Low-Dose Aspirin n = 38 (%)
Second-trimester loss	26 (42)	2 (5)
Perinatal deaths	28 (45)	2 (5)
IUGR	49 (79)	11 (29)
Survivors	8 (13)	34 (90)

* Excludes first trimester losses.
Adapted from Elder MG, deSwiet M, Robertson A, et al: Low-dose aspirin in pregnancy. Lancet i:410, 1988; with permission.

SAFETY OF THE USE OF LOW-DOSE ASPIRIN IN PREGNANCY

There are several reports describing an association between prenatal aspirin ingestion and adverse effects in the mother, fetus, and newborn. Most of these effects relate to inhibition of prostaglandin synthesis by pharmacologic doses of aspirin.⁴ Reported maternal side effects have included anemia, antepartum and postpartum hemorrhage, prolonged bleeding time, and prolonged gestation and prolonged duration of labor.⁸ Reported fetal effects have included congenital cardiac defects, oligohydramnios, fetal growth retardation, and increased frequency of stillbirths. Neonatal complications have included neonatal bleeding disorders and persistent pulmonary hypertension; however, the data from the Perinatal Collaborative Project,¹⁹ which included outcomes of 50,282 pregnancies, found no differences in congenital malformations, perinatal deaths, or neonatal birth weights between women taking aspirin and those not taking aspirin during pregnancy.

Several recent reports investigated the effects of low-dose aspirin on neonatal prostanoids, bleeding time, pulmonary circulation, and bleeding disorders. Ylikorkala et al²⁹ studied the effects of maternal ingestion of a single dose of 100 mg aspirin during labor on fetoplacental and neonatal prostanoid production. They found that this dose of aspirin inhibited only fetoplacental TXA₂ production without affecting neonatal PGI₂ production. Sibai et al¹⁸ studied the serum levels of 6-keto-PGF₁ α and neonatal TXB as well as neonatal pulmonary circulation in 40 neonates after maternal treatment with either placebo (n = 10), 20 mg aspirin/d (n = 10), 60 mg aspirin/d (n = 10), and 80 mg aspirin/d (n = 10) for 2 to 3 weeks. Neonatal serum levels of prostanoids studied were similar among the four study groups. All neonates had echocardiographic evidence of patent ductus arteriosus and none of the studied infants had evidence of clinical bleeding. Similar findings were recently reported by Viinikka et al.²³

To date, several studies have described the use of low-doses of aspirin in over 500 pregnancies. These studies reported no adverse maternal or neonatal side effects. It is interesting to note that closure of the ductus arteriosus and neonatal bleeding disorders were not present.

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